CH 9 — Critical Points

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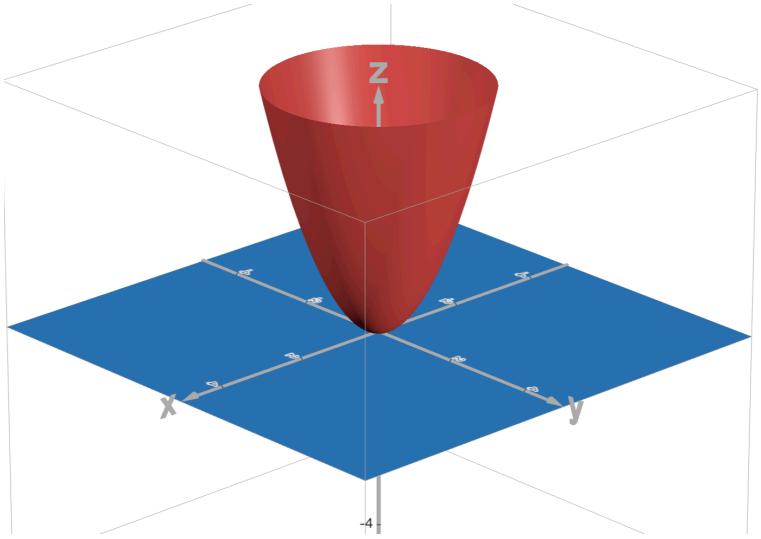
9.1 Critical Points and Local Extrema

Recall: In single -variable calculus, local extrema occur when $f'(x) = 0$ or when $f'(x)$ does not exist

> 📘 **Definition — Local Extrema**
>
> In two dimensions, (a, b) is a
>
> - **Local Maximum** of f if $f(x, y) \leq f(a, b)$ in some neighborhood of $f(a, b)$
> - **Local Minimum** of f if $f(x, y) \geq f(a, b)$ in some neighborhood of $f(a, b)$

Examples:

Local Minimum:



> 📐 **Theorem — Local Extrema**
>
> If $f(a, b)$ is a local extrema of $f(x, y)$, then
>
> $$f_x(a, b) = 0 \text{ or DNE}$$
>
> and
>
> $$f_y(a, b) = 0 \text{ or DNE}$$

Proof

Consider $g(x) = f(x, b)$. If (a, b) is a local extrema of f, $x = a$ must be a local extrema of $g(x)$. Therefore, $g'(a) = 0 \text{ or DNE} = f_x(a, b)$.

Similar argument can be applied for $g(x) = f(a, y)$

□

Definition — Critical Points

A point $(a, b) \in \text{dom} f$ is a critical point of f if

$$f_x(a, b) = 0 \text{ or DNE}$$

and

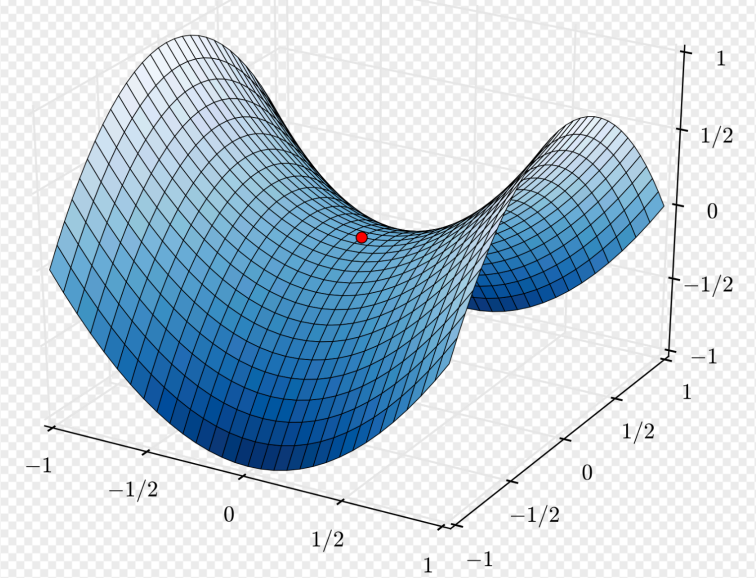
$$f_{y(a,b)} = 0 \text{ or DNE}$$

A local extrema must occur at a critical point BUT the reverse is not true.

Definition — Saddle Point

A critical point (a, b) is a **saddle point** of $f(a, b)$ if in every neighborhood of $(a, b) \exists (x_1, y_1)$ and (x_2, y_2) s.t.

$$f(x_1, y_1) > f(a, b) > f(x_2, y_2)$$



Example:

Find all critical points of $f(x, y) = x^2y - 2xy^2 + 3xy + 4$

$f_x = 2xy - 2y^2 + 3y, f_y = x^2 + 4xy + 3x$. It is two continuous functions, so the critical points must have $f_x = f_y = 0$

$2xy - 2y^2 + 3y = x^2 - 4xy + 3x = y(2x - 2y + 3) = x(x - 4y + 3)$

$y_1 = 0, x_1 = 0 : (0, 0)$

$y_2 = 0, x + 3 = 0 \implies x_2 = -3 : (-3, 0)$

$2x + 2y + 3 = 0, x_3 = 0 \implies -2y + 3 = 0, y_3 = \frac{3}{2} : \left(0, \frac{3}{2}\right)$

$2x - 2y + 3 = 0, x - 4y + 3 = 0 \implies x_4 = -1, y_4 = \frac{1}{2} : \left(-1, \frac{1}{2}\right)$

We have 4 critical points: $(0, 0), (-3, 0), \left(0, \frac{3}{2}\right), \left(-1, \frac{1}{2}\right)$

9.2 Second Derivative Test

Let $f(x,y)$ be a C^2 function in some neighborhood of a critical point (a,b), i.e. for (x,y) near (a,b),

$$f(x,y) \approx P_{2,(a,b)}(x,y)$$

Since (a,b) is a critical point and $f \in C^2$, then we must have $f_x(a,b) = 0 = f_y(a,b)$

$\therefore f(a,b) \approx f(a,b) + \frac{1}{2}\big(f_{xx}(a,b)(x-a)^2 + 2f_{xy}(x-a)(y-b) + f_{yy}(y-b)^2\big)$

$\implies f(x,y) - f(a,b) \approx \frac{1}{2}\big(f_{xx}(a,b)(x-a)^2 + 2f_{xy}(x-a)(y-b) + f_{yy}(y-b)^2\big)$

Let $A = f_{xx}(a,b), B = f_{xy}(a,b), C = f_{yy}(a,b), u = x-a, v = y-b$

$\implies \frac{1}{2}\left[Au^2 + 2Buv + Cv^2\right]$ is the quadratic form

$\implies \frac{1}{2}\left[A\left(u + \frac{Bv}{A}\right)^2 + \left(C - \frac{B^2}{A}\right)v^2\right]$

Discovery:

Case 1: $A > 0$ and $C - \frac{B^2}{A} > 0 \implies AC - B^2 > 0$ which is the local minimum

Case 2: $A < 0$ and $C - \frac{B^2}{A} < 0 \implies AC - B^2 > 0$ which is the local maximum

Case 3: A and $C - \frac{B^2}{A}$ has opposite signs $\implies AC - B^2 < 0$ then it could be a saddle point

Theorem — Second Derivative Test

Let $Hf = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$ be the Hessian matrix of $f \in C^2$ in some neighborhood of a critical point (a,b) and that

$$f_x(a,b) = f_y(a,b) = 0$$

1. $f_{xx}(a,b) > 0$ and $\det Hf(a,b) > 0$ or $Hf(a,b)$ satisfy positive definite, then (a,b) is local minimum
2. $f_{xx}(a,b) < 0$ and $\det\ Hf(a,b) > 0$ or $Hf(a,b)$ satisfy negative definite, then (a,b) is a local maximum
3. $\det Hf(a,b) < 0$, then (a,b) is a saddle point
4. $\det Hf(a,b) = 0$, inconclusive, or the degenerate critical point

Example:

$f(x,y) = 2(x-y)^2 - x^4 - y^4$ has critical points $\left(\sqrt{2}, -\sqrt{2}\right), \left(-\sqrt{2}, \sqrt{2}\right), (0,0)$

$f_x = 4(x-y) - 4x^3, f_y = 4(x-y) - 4y^3$

$f_{xx} = 4 - 12x^2, f_{xy} = f_{yx} = -4, f_{yy} = 4 - 12y^2$

$\det\left(Hf\left(\sqrt{2}, -\sqrt{2}\right)\right) = \det\left(\begin{bmatrix} -20 & -4 \\ -4 & -20 \end{bmatrix}\right) > 0, -20 > 0 \implies$ local maximum

$\det\left(Hf\left(-\sqrt{2}, \sqrt{2}\right)\right) = \det\left(\begin{bmatrix} -20 & -4 \\ -4 & -20 \end{bmatrix}\right) > 0, -20 > 0 \implies$ local maximum

$\det(Hf(0,0)) = \det\left(\begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix}\right) = 0 \implies$ inconclusive.

We will investigate the sign $f(x,y) - f(0,0) = 2(x-y)^2 - x^4 - y^4 - 0$

If $y = 0$, $f(x,0) - f(0,0) = 2(x)^2 - x^4 = x^2(2 - x^2) > 0,\ 0 < x^2 < 2$

If $y = x$, $f(x,x) - f(0,0) = -2x^4 < 0,\ \forall x \neq 0$. Thus it is saddle point from the above discovery.

9.3 Proof of the Second Partial Derivative Test